

DanceMimic: Awaken Your Dancing Instinct through a Real-time Dance Imitation Capture System

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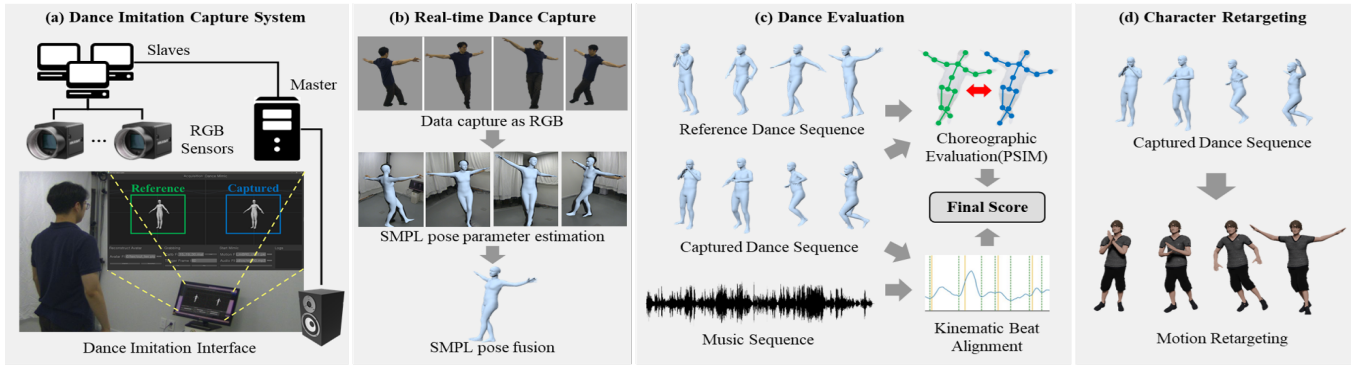


Figure 1: Overall pipeline and features of DanceMimic.

Abstract

In this demonstration, we present DanceMimic, a real-time dance imitation capture and assessment system designed to enhance the accessibility and learning experience of dancing. Guided by an interactive user interface, novice dancers can simultaneously observe and imitate a selected choreography while listening to the corresponding music. The choreography is captured and compared with the reference for quantitative evaluation of the performance proficiency. Finally, our system retargets the performed dance to a rigged 3D character to provide immersive imitation experience. Demo video is on: <https://youtu.be/7nL9YPPRj-4>

CCS Concepts

• **Computer systems organization** → **Real-time operating systems**; • **Networks** → **Error detection and error correction**.

Keywords

Real-time Capture, Dance Assessment, Motion Retargeting, Animatable Character

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1 Introduction

Dance, a natural human response to music, is one of the most common methods to express and discharge one's emotions. While people can dance instinctively, proper learning is essential to perform choreography that matches the beat and best expresses the mood of the music. However, the process of learning is highly intricate without professional instruction, which requires significant time and financial investment. To alleviate this issue, we present DanceMimic, a real-time dance imitation capture and assessment framework that allows individuals to easily learn by themselves. Using multiple synchronized RGB cameras, the dance performance is captured in real-time without any markers or specialized suits. Through an interactive user interface, the target choreography and corresponding music are played along with the individual's captured performance, fostering real-time feedback. After the performance is finished, a final assessment score is provided. As choreographic correctness and beat matching are key to dance assessment, we integrate a pose similarity metric [8] and kinematic beat alignment metric [2–4] to compute the final score. To provide an immersive imitation experience, our system retargets the captured choreography to a 3D character. Through DanceMimic, a user can intuitively and independently learn how to dance without requiring professional feedback.

2 System Framework

2.1 Dance Imitation Capture System

The architecture of the dance imitation system of DanceMimic is depicted in Figure 1-(a). A total of four slave PCs are connected to a single master PC. Each slave PC reads frames acquired from a single RGB camera at 60 FPS. All four cameras are synchronized through a hardware trigger generator. The frames are processed on the slave PCs, and only the results are sent to the master PC. On the master PC, we implement an intuitive dance imitation interface for dancers to easily imitate a target choreography.

2.2 Real-time Dance Capture

In order to capture the dance performance in a markerless setting, we employ a simple yet effective multi-view 3D pose estimation algorithm. We first compute the intrinsic and extrinsic camera parameters by calibrating all cameras using a checkerboard. Given multi-view RGB images, we then estimate the pose of the dancer by extracting the parameters of the SMPL [11] model for each viewpoint. SMPL is a parametric human body model where the pose and shape parameters represent the rotations of each joint and the body shape, respectively. We use the CLIFF [10] model to extract the SMPL parameters, which is accelerated by the NVIDIA TensorRT [13] engine to ensure real-time performance. To compensate for self-occlusion in monocular viewpoints, we fuse the SMPL pose and shape parameters across multiple viewpoints after transforming them to the world coordinate system using the camera parameters. The fused SMPL parameters are obtained by taking the average of the transformed parameters, specifically by applying Singular Value Decomposition (SVD) [1] on the stacked parameter matrix.

2.3 Dance Assessment

As shown in Figure 1-(c), we evaluate the captured choreography performance from two perspectives: choreographic accuracy and kinematic beat alignment. Choreographic accuracy is measured using the pose similarity metric (PSIM) proposed by [8]. This method considers local and global structural differences of the motion, along with the joint-wise differences. The PSIM metric is applied frame-by-frame on the captured choreography and the reference choreography to derive the score. The PSIM scores are temporally pooled to provide the final choreographic accuracy score. On the other hand, kinematic beat alignment measures the consistency between the kinematic features from the motion and the musical beats from the audio. Following the work of [4], kinematic features are extracted by finding the local minima of the kinematic velocity, and music beats are extracted using Librosa [12]. The kinematic beat alignment score is then computed by measuring the average distance between the motion beats and the nearest music beats. Finally, we predict the final score of the performance by applying linear regression on the two scores.

3 Experiment Results

To verify the effectiveness of DanceMimic, we conducted experiments on real-time dance capture and dance assessment. We obtained paired data from an OptiTrack MoCap system [14] and our

Table 1: Evaluation of our 3D pose estimation method

Method	MPJPE(↓)	PA-MPJPE(↓)
Monocular	107.1	60.9
Multi-view Fusion	93.8	52.3

Table 2: Comparison with other assessment methods

Method	PLCC(↑)	SRCC(↑)
PA-MPJPE	0.7244	0.7311
PSIM [8]	0.7807	0.7949
DanceQA [2]	0.8123	0.8104
DanceMimic (Ours)	0.8052	0.8079

dance imitation capture system simultaneously through hardware synchronization. We captured three subjects imitating ten choreography sequences sampled from the AIST++ dataset [9]. A total of five professional dancers took part in the subjective assessment of the imitated dance sequences, evaluating how well the captured sequences represented the original sequences. For dance capture evaluation, we measured the average 3D joint error in millimeters between the motion captured from our system and the OptiTrack system. For dance assessment, we follow the validation strategy of [5–7] and evaluate the correlation of the assessment performance of each method using Pearson’s linear correlation coefficient (PLCC) and Spearman’s rank-order correlation coefficient (SRCC). The results for dance capture and assessment are shown in Table 1 and Table 2. As shown in Table 1, our capture method managed to effectively solve self-occlusion through multi-view fusion. Despite its simplicity, as shown in Table 2, our assessment method showed performance competitive to existing assessment methods.

4 User Experience

A dancer selects a choreography and corresponding background music through the imitation interface, and the imitation process starts after a three-second countdown. The interface displays the reference choreography on the left-hand side and the captured choreography on the right-hand side. The corresponding background music is played through a speaker simultaneously. After the dance performance is finished, the final assessment score is provided to give the dancer a sense of the overall performance quality. During the motion capture process, to provide a better imitation experience, we retarget the captured motion of the dancer to a 3D character, as shown in Figure 1-(d). The retargeted motion of the character helps the dancer monitor their performance while providing an immersive experience. This feature enhances DanceMimic by offering a comprehensive and user-friendly tool for learning and perfecting dance routines independently.

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